

Algebra II with Trigonometry

Chapter 11.1

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

- 1. Circle Basics** A circle is the set of all points equidistant from a fixed point. What is that fixed point called?
- 2. Circle at the Origin** What is the equation of a circle centered at the origin with radius 5?
- 3. Center and Radius** For the circle $(x - 2)^2 + (y + 3)^2 = 16$, what are the center and radius?
- 4. Parabola Direction** If $y = ax^2 + bx + c$ and $a < 0$, does the parabola open upward or downward?
- 5. Vertical Parabola Form** A parabola has equation $x^2 = 12y$. Does it open up or down, and what is the value of p ?
- 6. Horizontal Parabola Form** A parabola has equation $y^2 = -20x$. Does it open left or right, and what is the value of p ?
- 7. Focus and Directrix** For the parabola $x^2 = 8y$, what are the focus and directrix?
- 8. Quick Rewrite** Rewrite $5x + 3y^2 = 0$ into standard parabola form, then state whether it opens left or right.
- 9. Build the Equation** What is the standard equation of the parabola with vertex at the origin and focus $(0, -4)$?

10. **Distance Idea** A parabola has vertex at the origin and directrix $y = -3$. What is its focus, and does it open upward or downward?